School of Future Tech

Case Study Report

on

HealthMate: Preventive Health Tracker

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Course:	HTML5, CSS3, JavaScript
Program:	B.Tech CSE
University:	ITM Skills University
Semester:	II
Year:	2026

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1. Introduction to the Case Study

In today's fast-paced world, people often neglect monitoring their basic health indicators such as Body Mass Index (BMI), blood pressure, and sugar levels. This negligence can lead to serious health complications that could have been prevented with regular tracking and awareness.

This case study focuses on the design and development of **HealthMate**, a preventive health tracking web application. HealthMate enables users to record health metrics, calculate BMI automatically, view health summaries, and access doctor consultation information — all through a responsive, user-friendly interface built entirely with front-end web technologies.

The project demonstrates practical implementation of HTML5, CSS3, and JavaScript to create a functional multi-page health application. It highlights how structured form handling, dynamic calculations, and local data storage can be combined to deliver real-world value.

2. Problem Statement / Case Background (Abstract)

Background

Regular monitoring of health parameters such as BMI, blood pressure, and sugar levels is critical for preventive healthcare. However, most people lack a simple, accessible tool to consistently track and review their health data. Existing health applications are often overly complex or require expensive subscriptions.

Abstract

This case study presents the design and implementation of HealthMate, a preventive health tracking web application developed using HTML5, CSS3, and JavaScript. The system allows users to input health data including height, weight, blood pressure (systolic and diastolic), and sugar levels. BMI is calculated automatically using the standard WHO formula. Health records are stored persistently using the browser's LocalStorage API, and a dedicated reports dashboard provides an at-a-glance health summary. The application also includes a doctor consultation section and a subscription plans page.

3. Case Study Design

The HealthMate system is designed as a modular multi-page web application with five pages, each serving a distinct role in the health monitoring workflow:

Home Page (index.html)

Serves as the landing page. Displays the application title, a 'Start Tracking' CTA button, and four feature cards — BMI Calculator, Health Tracking, Health Reports, and Doctor Consultation.

Tracker Page (tracker.html)

The core input interface. Users enter date, height, weight, BP systolic, BP diastolic, and sugar level. Two action buttons allow BMI calculation and saving the record to

LocalStorage. A data table below displays all saved entries.

Reports Page (reports.html)

Presents a Health Summary dashboard with four metric cards: Average BMI, Blood Pressure, Sugar Level, and overall Health Status with a colour-coded verdict (e.g., 'Healthy' in green).

Doctor Page (doctor.html)

Lists doctor profiles (Dr. Mehta – Cardiologist, Dr. Sharma – General Physician, Dr. Patel – Nutritionist) with Book consultation buttons.

Plans Page (plans.html)

Displays three subscription tiers — Basic (Free), Pro (■299/month), and Premium (■599/month) — with feature comparison and call-to-action buttons.

System Architecture Summary

Component	Technology	Purpose
Structure	HTML5	Page layout, forms, navigation
Styling	CSS3 (Grid + Flexbox)	Responsive design, visual design
Logic	JavaScript (ES6)	BMI calc, DOM updates, form handling
Storage	LocalStorage API	Persistent health record storage

4. Methods & Algorithms / Technology Applied

HTML5

HTML5 defines the structural foundation of all pages. Semantic tags like <header>, <nav>, <section>, and <footer> improve readability and accessibility. Forms with appropriate input types (date, number, text) provide a clean data entry experience.

CSS3

CSS3 handles all visual presentation. CSS Grid and Flexbox create the responsive card-based layouts seen across all pages. A consistent maroon (#8B2346) colour palette is applied via CSS variables, with box-shadows, border-radius, and hover transitions enhancing the UI.

JavaScript (ES6+)

Key JavaScript functionalities implemented:

- BMI Calculation: Weight (kg) ÷ Height (m)^2 with automatic BMI category classification
- Event listeners on 'Calculate BMI' and 'Save Record' buttons for real-time feedback
- Dynamic table row generation to display saved health records
- LocalStorage read/write operations for data persistence across sessions
- DOM manipulation to update health summary values on the Reports page

LocalStorage API

Each health record is serialized to JSON and stored in the browser's LocalStorage. This allows data to persist after page refresh. The Reports page retrieves and parses this data to compute averages and determine overall health status.

5. Implementation Details and Snapshots

Form Handling (Tracker Page)

The Tracker page presents a clean centered card with input fields for Date, Height (cm), Weight (kg), BP Systolic, BP Diastolic, and Sugar Level. On clicking 'Calculate BMI', the value is computed and shown instantly. 'Save Record' writes the entry to LocalStorage and appends a new row to the records table below.

BMI Calculation Logic

$$\text{BMI} = \text{Weight (kg)} \div (\text{Height (m)})^2$$

Height is converted from cm to metres before the calculation. The result is classified: Underweight (<18.5), Normal (18.5–24.9), Overweight (25–29.9), Obese (≥30).

Application Screenshots

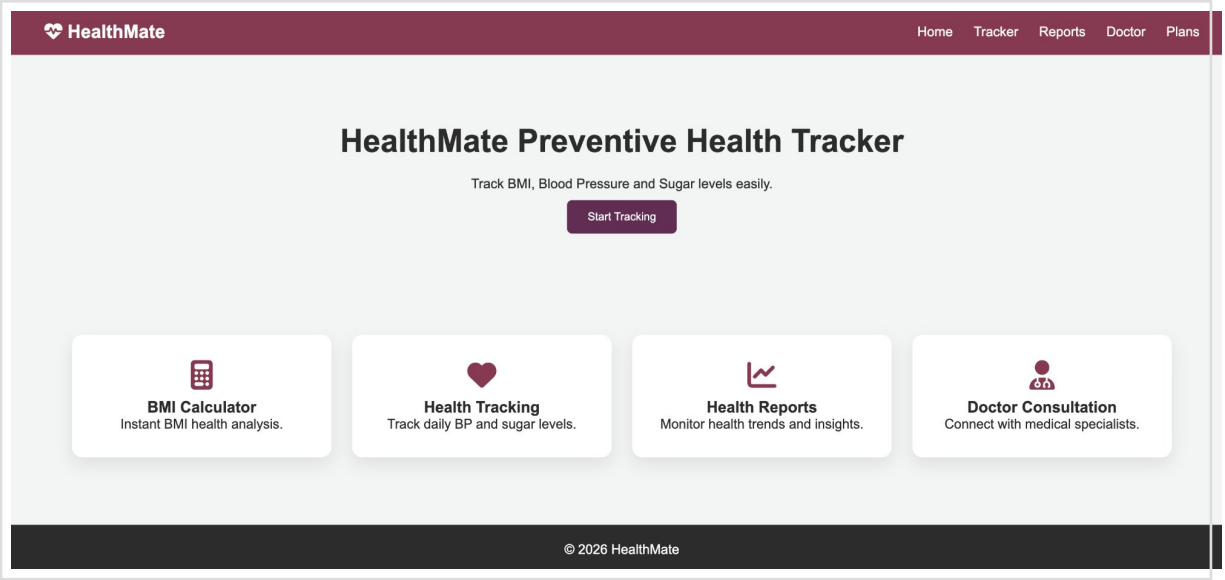

Figure 1: Home Page — Feature overview and Start Tracking CTA

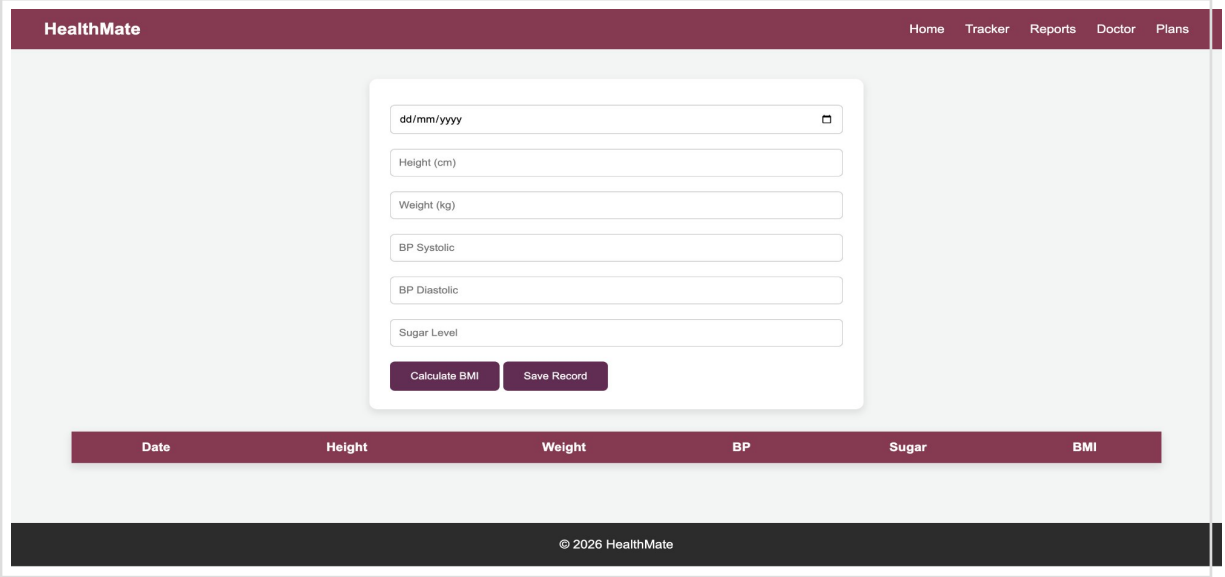

Figure 2: Health Tracker Page — Data entry form and records table

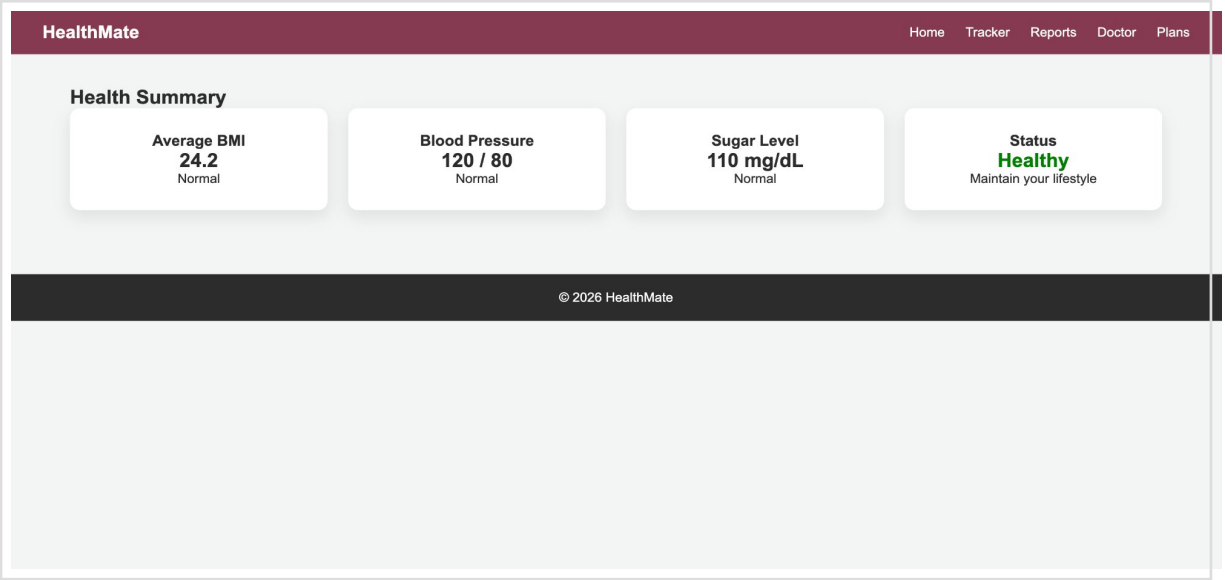

Figure 3: Health Reports Page — Summary dashboard showing BMI, BP, Sugar & Status

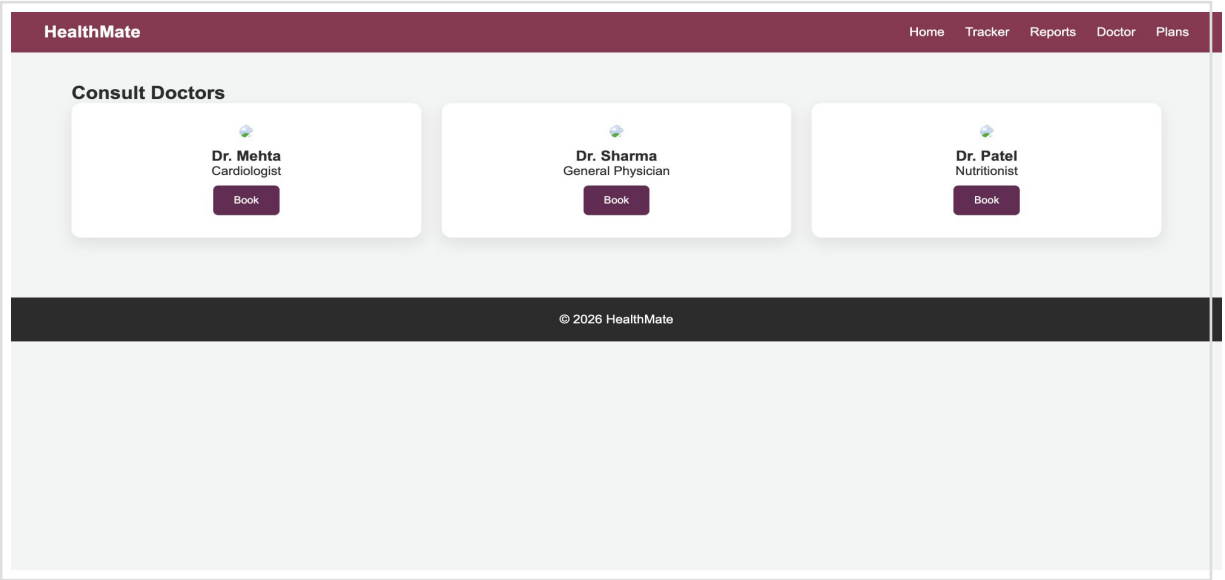

Figure 4: Doctor Consultation Page — Doctor profiles with Book buttons

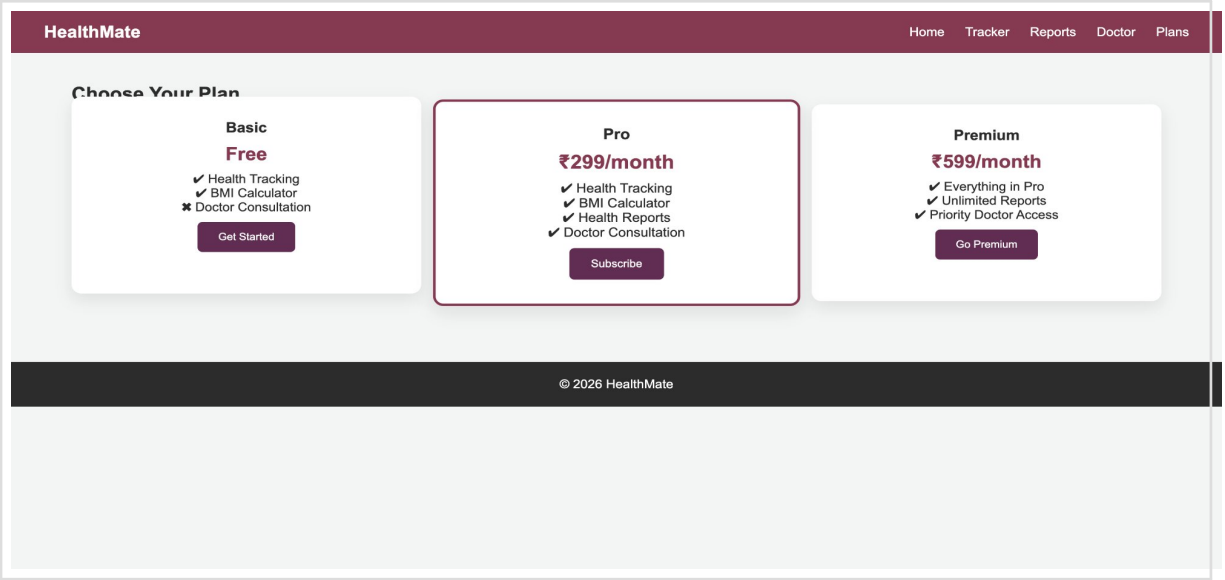

Figure 5: Subscription Plans Page — Basic, Pro, and Premium tier comparison

6. Results and Conclusion

Findings

- BMI is computed and displayed correctly for all valid user inputs, with accurate category classification.
- Health records persist across browser sessions via LocalStorage, confirming reliable client-side storage.
- The Reports dashboard correctly aggregates saved data and renders an intuitive health summary.
- The responsive layout renders cleanly across desktop, tablet, and mobile screen sizes.
- The doctor consultation and subscription plan pages provide a complete, real-world application feel.

Conclusion

The HealthMate Preventive Health Tracker successfully demonstrates the practical application of front-end web development technologies in solving a real-world healthcare challenge. The system integrates HTML5, CSS3, and JavaScript into a coherent, user-friendly multi-page platform that fulfills all defined objectives — from automatic BMI calculation to persistent health record storage and intuitive data visualization.

The project serves as a strong foundation for a production-grade health monitoring application and highlights how front-end technologies alone can deliver meaningful value to end users.

Future Scope

- User authentication and personalized login system
- Integration with wearable health devices (smartwatches, fitness bands)
- Cloud-based health record storage for cross-device access
- AI-based personalized health recommendations
- Mobile application development (Android/iOS)

7. References

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Acknowledgement

I would like to express my sincere gratitude to my faculty and ITM Skills University for providing the opportunity to work on this project. Their guidance throughout the course helped in successfully completing this case study. I also acknowledge the open-source community and documentation resources that greatly assisted in the development process.